
Averaged Jacobian Regularity Does Not Order Few-Step Error of Gaussian Interpolant ODEs

Anonymous Author(s)

Affiliation

Address

email

Abstract

1 Few-step sampling of a continuous-time generative model requires a probability
2 path, a solver, and a budget of field evaluations (NFE). Chen et al. propose mini-
3 mizing averaged squared Jacobian regularity A_2 as a path-selection criterion. We
4 ask whether the order that this scalar induces between two interpolants is reliable
5 at fixed NFE. Already for independent $N(0, 1)$ and $N(0, 4)$, the exact integrals are
6 $5\pi/8 - 1$ (linear) and $\pi^2/16$ (trigonometric VP), while Heun at NFE 8 reverses
7 Gaussian W_2 . The linear factor is the rational $6797469/3559400$; a rational enclo-
8 sure in $\mathbb{Q}[\pi, \sqrt{2}]$ gives $r_{VP} < 187/100$. On four centered commuting geometries,
9 the per-mode log-covariance schedule (Chen et al. Ex. 3.3), which minimizes $\mathcal{R}$,
10 attains the smallest W_2 in all 36 solver-budget blocks. Linear versus VP invert in
11 5 of 12 geometry $\times$ solver cells (4 of 12 at every NFE in $\{8, 16, 32\}$), from three
12 distinct $\mathcal{R}$ comparisons. Thus $\mathcal{R}$ is reliable as an objective on this family and not
13 reliable as a linear-versus-VP comparator away from that optimum.

14 1 Introduction

15 Few-step sampling commits to a path, a solver, and a small NFE budget before any error is observed.
16 Lipschitz constants enter classical one-step bounds [Hairer et al., 1993]. Chen et al. [2025] propose
17 minimizing averaged squared drift Lipschitzness (their Definition 3.2, denoted A_2) as a *selection*
18 criterion. They prove no bound relating A_2 to a discretization error. We test whether the pairwise
19 order that A_2 induces between two admissible paths is reliable at equal NFE.

20 We isolate that ranking question on commuting Gaussian interpolants, where the field is affine,
21 Gaussian W_2 is closed-form, and the regularity integrand is known exactly. The leading result is a
22 deterministic scalar Heun example (Section 3). A modal identity then records the signed information
23 that $\mathcal{R}$ discards (Section 4). A third path—the Cauchy–Schwarz minimizer of $\mathcal{R}$ —and a λ -sweep
24 show that $\mathcal{R}$ is a reliable *objective* on this family and an unreliable linear-versus-VP *comparator*
25 (Section 5).

26 Contemporaneous work studies Lipschitz Euler rates [Stéphanovitch, 2026], sharpness-aware
27 grids [Gupta et al., 2026], least-action paths [Du et al., 2026], and instance-aware discretizations [Yuan
28 et al., 2026]. Those results control other objects. Tao and Choi [2026] give the general strain/vorticity
29 form of the phenomenon; Section 6 records the precise relation.

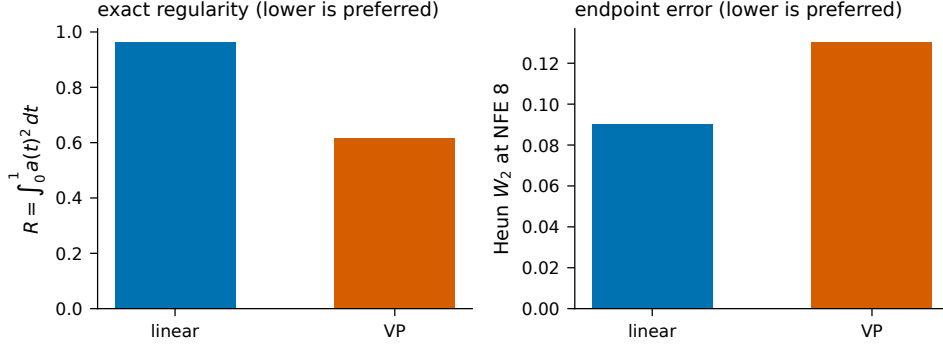


Figure 1: Proposition 1: exact $\mathcal{R}$ versus certified Heun W_2 at NFE 8. The VP bar is the certified lower bound.

2 Gaussian flows and averaged regularity

Let $X_0 \sim N(\mu_0, \Sigma_0)$ and $X_1 \sim N(\mu_1, \Sigma_1)$ be independent, and let (α, σ) satisfy $\alpha(0) = \sigma(1) = 1$ and $\alpha(1) = \sigma(0) = 0$. The interpolant $X_t = \alpha X_0 + \sigma X_1$ [Albergo et al., 2025, Lipman et al., 2023] is Gaussian with mean $m_t = \alpha\mu_0 + \sigma\mu_1$ and covariance $Q = \alpha^2\Sigma_0 + \sigma^2\Sigma_1$. The marginal field is affine,

$$b(t, x) = A(t)x + c(t), \quad A = C_t Q^{-1}, \quad c = \dot{m} - Am, \quad (1)$$

with $C_t = \dot{\alpha}\Sigma_0 + \dot{\sigma}\Sigma_1$. If $\Sigma_0 = I$, then Q, Q', A , and Σ_1 share the eigenbasis of Σ_1 , and mode i obeys $x'_i = a_i(t)x_i + c_i(t)$ with $q_i = \alpha^2 + \lambda_i\sigma^2$ and $a_i = q'_i/(2q_i)$. The covariance drift formula is independent of the mean path $c(t)$. We compare the linear path $(1-t, t)$ and trigonometric VP $(\cos(\pi t/2), \sin(\pi t/2))$. Following Chen et al. [2025, Definition 3.2], $\mathcal{R}[b] = \int_0^1 \|A(t)\|_2^2 dt$. For state-independent A this coincides with their path expectation. By definition $\mathcal{R}$ ignores $c(t)$. Error is Gaussian W_2 of the endpoint laws [Gelbrich, 1990]. This is neither pathwise trajectory error nor local truncation error. Equal NFE uses $S = B/s$ steps with $s = 1, 2, 4$ for Euler, Heun, and RK4.

Any admissible variance path from 1 to $\lambda > 0$ obeys $\int_0^1 a dt = \frac{1}{2} \ln \lambda$. Hence $\mathcal{R} = \int_0^1 a^2 dt = (\int a)^2 + \text{Var}_t(a) \geq (\frac{1}{2} \ln \lambda)^2$, with equality if and only if a is constant. The minimizer is $q(t) = \lambda^t$, $a \equiv \frac{1}{2} \ln \lambda$, which is Chen et al. Example 3.3. In several dimensions, the per-mode schedule $q_i(t) = \lambda_i^t$ yields $\mathcal{R} = (\frac{1}{2} \max_i |\ln \lambda_i|)^2$. It is not in general a shared (α, σ) interpolant.

3 A one-dimensional Gaussian ranking counterexample

Assumption 1. Endpoints $N(0, 1)$ and $N(0, 4)$ are independent. Schedules are linear and trigonometric VP. The solver is explicit Heun with NFE 8 (four steps of size $1/4$).

Proposition 1. Under the assumption, $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1 > \mathcal{R}_{\text{VP}} = \pi^2/16$, while the Heun factors satisfy $r_{\text{lin}} = 6797469/3559400$ and $0 < r_{\text{VP}} < 187/100$. Hence $W_2^{\text{lin}} < 0.091 < 0.130 < W_2^{\text{VP}}$.

Proof sketch. The linear variance is $q_{\text{lin}}(t) = 1 - 2t + 5t^2$. Completing the square gives the equivalent identities $q_{\text{lin}}(t) = 5[(t - 1/5)^2 + (2/5)^2] = 5(t - 1/5)^2 + 4/5$. The substitution $u = t - 1/5 = (2/5) \tan \theta$ converts $\int a_{\text{lin}}^2$ into $(5/2) \int \sin^2 \theta d\theta$. Limits $\arctan(-1/2)$ and $\arctan 2$ together with $\arctan 2 + \arctan(1/2) = \pi/2$ yield $\mathcal{R}_{\text{lin}} = 5\pi/8 - 1$. The VP integral evaluates to $\pi^2/16$ by a Weierstrass identity. The four linear Heun factors are the rationals $47/52, 6/5, 497/370$, and $97/74$, all positive. Exact VP grid values are $a(0) = a(1) = 0, a(1/2) = 3\pi/10$, and $a(1/4) = (9 + 15\sqrt{2})\pi/82, a(3/4) = (-9 + 15\sqrt{2})\pi/82$. Positivity of $a(3/4)$ is $15^2 \cdot 2 - 9^2 = 369 > 0$. The four-step product lies in $\mathbb{Q}[\pi, \sqrt{2}]$ with nonnegative coefficients, so substituting $\sqrt{2} < 99/70$ and $\pi < 355/113$ is valid and yields $r_{\text{VP}} < 187/100$ by an integer comparison in the anonymous supplement. Interval arithmetic is a cross-check only. $\square$

Table 1: Per-mode log-covariance (Chen Ex. 3.3) versus linear and VP. $\mathcal{R}$ is continuous; the last column is the W_2 -minimizer at Heun, NFE 8. The same W_2 preference holds in all 36 blocks.

family	d	$\mathcal{R}_{\text{lin}}$	$\mathcal{R}_{\text{VP}}$	$\mathcal{R}_{\text{log}}$	Heun-8 W_2 min
anisotropic	2	0.972	0.192	0.120	log-cov
anisotropic	8	0.972	0.192	0.120	log-cov
low-rank	2	1.246	0.625	0.380	log-cov
low-rank	8	2.944	4.731	2.244	log-cov

Table 2: Strongest centered inversion: low-rank, $d = 8$, Euler, NFE 8. Eigenvalues: 0.05 (six times), 4.64827725629505, 10.675638226546441.

path	$\mathcal{R}$	$\widehat{\mathcal{R}}_{24}$	Gaussian W_2
linear	2.9441044083	2.9476523251	0.8108540111
VP	4.7305438136	4.7295206355	0.4564779075

4 Sign change and transported defects

For any independent linear coupling, $a_{\text{lin}}(0) = -1 < 0 < a_{\text{lin}}(1) = +1$. The VP drift vanishes at the endpoints and does not change sign on $(0, 1)$. Averaging a^2 therefore erases a qualitative distinction that every fixed-stage Runge–Kutta method samples.

Let $e_n = \sqrt{q(t_{n+1})/q(t_n)}$ and let r_n be the numerical factor. The telescoping identity

$$\prod_n r_n - \prod_n e_n = \sum_n (r_n - e_n) \left(\prod_{j < n} r_j \right) \left(\prod_{j > n} e_j \right) \quad (2)$$

is an accounting tool, not the content of the inversion. The content is the leading exact-minus-method coefficient: $\frac{1}{2}(a' + a^2)h^2$ for Euler and $(-a''/12 + a^3/6)h^3$ for the implemented Heun factor. Those expansions involve different derivatives, signs, and stage locations.

5 The $\mathcal{R}$ -minimizer and the inversion region

Source $N(0, I)$. Four centered targets: anisotropic Gaussians with condition number 4 in dimensions 2 and 8 (eigenvalues a geometric sequence from $1/2$ to 2); low-rank covariances $FF^\top + 0.05I$ of rank 2 (seed 271828). Because $\mathcal{R}$ uses the spectral norm, the two anisotropic dimensions share the same $\mathcal{R}$ pair (0.972296, 0.192270), so the grid contains three distinct linear-versus-VP $\mathcal{R}$ comparisons. Each geometry is run under Euler, Heun, and RK4 at NFE $\{8, 16, 32\}$. Headline $\mathcal{R}$ is the continuous integral of $\max_i a_i(t)^2$.

The per-mode log-covariance path $q_i(t) = \lambda_i^t$ has the smallest $\mathcal{R}$ on every geometry and the smallest Gaussian W_2 in all 36 blocks (Table 1 reports Heun at NFE 8). Linear versus VP invert in 5 of 12 geometry $\times$ solver cells; 4 of those 12 invert at every tested NFE. The fourteen inverted solver-budget rows are a hierarchical listing of those cells, not a population rate.

A one-dimensional sweep $\lambda \in [0.05, 100]$ (Figure 2) locates the linear-versus-VP inversions. At $\lambda = 4$, the inversion is present under Heun at every tested budget NFE $\in \{4, 8, 12, 16, 24, 32, 64, 128, 256\}$, absent under Euler, and mixed under RK4 (absent at NFE 8 and 12). Euler *does* invert at more extreme λ on the same sweep. The scalar example is therefore solver-dependent, not an artefact of Heun at budget 8 alone.

6 Related work and limitations

Flow matching and interpolants supply the Gaussian family [Lipman et al., 2023, Albergo et al., 2025, Liu et al., 2023, Tong et al., 2024]. Lipschitz-guided design proposes the selection criterion tested [Chen et al., 2025]. Solver-specific schedules and exact linear parts are documented in diffusion practice [Sabour et al., 2024, Karras et al., 2022, Lu et al., 2022, Zhang and Chen, 2023].

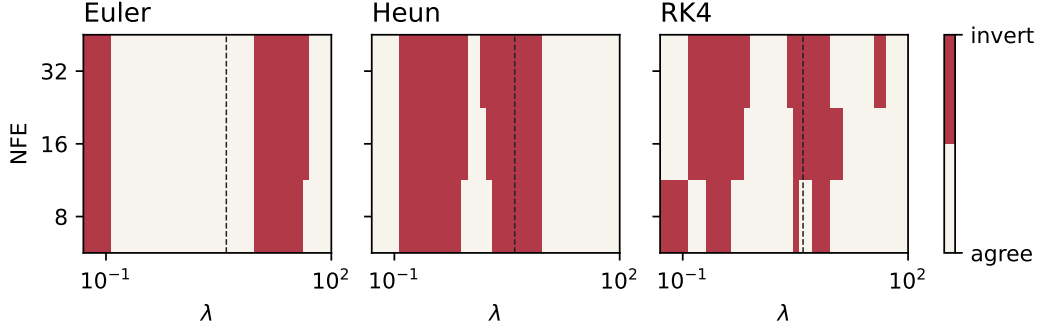


Figure 2: Linear versus VP ranking inversion versus target variance λ . Dark cells: $\mathcal{R}$ and W_2 disagree. The dashed line is $\lambda = 4$ (Proposition 1).

90 Stéphanovitch [2026] prove sharp Lipschitz regularity of canonical flow-matching fields and Euler-
 91 type W_2 rates; that is a bound, not a pairwise ranking of two interpolants at equal NFE.

92 Tao and Choi [2026] decompose $\nabla v = S + \Omega$ and control Euler error by the logarithmic norm
 93 $\mu_2 = \lambda_{\max}(S)$, which is signed, whereas $\mathcal{R}$ integrates $\|A\|_2^2$. In one dimension $\mu_2(a) = a$ and
 94 $\|A\|_2 = |a|$. Their Theorem 5 states that McCann displacement interpolation has vanishing material
 95 derivative, hence Euler is order 2 in the Eulerian analysis; Remark 7 records the stronger Lagrangian
 96 exactness observed on that interpolant. Neither statement applies to the independent-coupling linear
 97 or VP path studied here. The present paper supplies a rationally certified Gaussian instance of ranking
 98 failure for the unsigned average $\|A\|_2^2$. A counterexample to μ_2 itself is outside our scope.

99 The systems are commuting affine Gaussian interpolants, three fixed-step Runge–Kutta methods, and
 100 small equal-NFE budgets. No learned neural field is evaluated. Chen et al. prove no discretization-
 101 error bound and do not claim a universal pairwise ranking at fixed NFE; the example does not
 102 invalidate A_2 as an optimization objective within a path family. The Gaussian family is a probe
 103 in which $\mathcal{R}$ and W_2 are closed form; in a learned field the linear part is not exact, so exponential
 104 integrators that solve an affine scalar mode exactly are not a drop-in substitute. Cell counts are
 105 hierarchical descriptive summaries, not population rates.

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